

XAI-TransMed: Explainable AI Transformer-Based Medical Model for Alzheimer's Disease Detection and Progression Rate

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COMPUTER SCIENCE AND ENGINEERING

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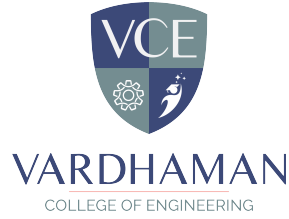
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Abstract

Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking ability, and daily functioning. Early detection and accurate prediction of disease progression are important for effective treatment and better patient care. In recent years, artificial intelligence and deep learning techniques have been widely used in medical imaging, especially with MRI scans, to support diagnosis. However, traditional models like Convolutional Neural Networks (CNNs) mainly focus on local features and fail to capture long-range relationships between different brain regions. In addition, these models lack transparency and are often considered “black-box” systems, making them less reliable for clinical use. To overcome these challenges, this paper proposes XAI-TransMed, a transformer-based framework that uses a Vision Transformer (ViT) with self-attention mechanisms to capture global relationships in brain imaging data and improve feature representation.

The proposed system also integrates multimodal data by combining MRI images with clinical and demographic information such as age and cognitive scores, which improves prediction accuracy and robustness. The model performs dementia stage classification into four categories: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia, and also predicts disease progression using machine learning techniques like XGBoost and Cox regression. A key feature of this work is the use of explainable AI techniques such as attention maps and Grad-CAM, which highlight important brain regions influencing the model’s predictions and improve interpretability. The framework is evaluated on the OASIS dataset and shows high accuracy, reliable performance, and efficient computation, making it suitable for clinical decision support. Overall, the proposed system provides an accurate, interpretable, and practical solution for early detection and monitoring of Alzheimer’s disease.

Keywords: Alzheimer’s Disease, Explainable Artificial Intelligence (XAI), Vision Transformer (ViT), Deep Learning, MRI, Medical Imaging, Disease Progression Prediction, Multimodal Data, Self-Attention Mechanism.

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Abbreviations

Abbreviation	Description
AD	Alzheimer's Disease
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
ViT	Vision Transformer
MRI	Magnetic Resonance Imaging
MMSE	Mini-Mental State Examination
XGBoost	Extreme Gradient Boosting
LSTM	Long Short-Term Memory
OASIS	Open Access Series of Imaging Studies
ML	Machine Learning
DL	Deep Learning

CHAPTER 1

Introduction

1.1 Introduction

Alzheimer’s disease is a progressive neurodegenerative disorder that affects memory, thinking ability, and the capacity to perform daily activities. It is one of the leading causes of cognitive decline among the elderly population worldwide and poses a serious challenge to healthcare systems. As the disease progresses, patients experience severe memory loss, confusion, and difficulty in carrying out routine tasks, which significantly impacts their quality of life. Therefore, early detection and accurate prediction of disease progression are extremely important for timely treatment, proper care, and better management of the condition.

In recent years, artificial intelligence (AI) and deep learning techniques have gained significant attention in the field of medical imaging. These technologies have shown great potential in analyzing complex medical data such as MRI brain scans. By automatically extracting patterns and features from imaging data, AI-based systems can assist doctors in diagnosing diseases more accurately and efficiently. This has opened new possibilities for improving the early detection of Alzheimer’s disease and supporting clinical decision-making.

This project focuses on developing an advanced system called XAI-TransMed, which is a transformer-based framework designed for Alzheimer’s disease detection and progression prediction. Unlike traditional deep learning models such as Convolutional Neural Networks (CNNs), the proposed system uses Vision Transformers (ViT) that can capture global relationships within MRI images. This helps in better understanding of how different regions of the brain are affected. In addition, the system integrates Explainable Artificial Intelligence (XAI) techniques to make the predictions more transparent and understandable. By generating visual explanations such as heatmaps, the model highlights the

important brain regions that influence its decisions, thereby increasing trust and usability in clinical environments.

Another important aspect of this project is the integration of multimodal data, where both MRI imaging features and clinical information such as age, education level, and cognitive scores are used together. This provides a more comprehensive understanding of the patient's condition and improves the overall prediction accuracy. The system is capable of not only classifying different stages of dementia but also predicting disease progression over time.

The main aim of this project is to develop a system that improves the accuracy, reliability, and interpretability of Alzheimer's disease detection. By combining advanced deep learning techniques with explainability, the project ensures that the model's decisions are understandable to medical professionals. This makes the system more suitable for real-world applications and helps in bridging the gap between technology and healthcare.

This report is organized into multiple chapters, each focusing on different aspects of the project. It includes a detailed literature review, system analysis and design, implementation methodology, testing procedures, and discussion of results. Finally, the report concludes with key findings and suggestions for future improvements.

1.2 Background and Motivation

Alzheimer's disease has become a major healthcare challenge in recent years, mainly due to the rapid growth of the aging population across the world. As life expectancy increases, the number of people affected by cognitive disorders is also rising significantly. This has created an urgent need for effective diagnostic tools that can identify the disease at an early stage and help in managing its progression. Traditionally, Alzheimer's diagnosis depends on clinical evaluation, patient history, cognitive tests, and neuroimaging techniques such as MRI scans. While these methods are useful, they often require expert interpretation and may not always provide consistent or early-stage detection.

With the advancement of artificial intelligence, machine learning and deep learning techniques have been increasingly used in the medical field to analyze

complex data. In particular, these methods have shown great potential in processing medical images such as MRI scans, where they can automatically detect patterns that may not be easily visible to the human eye. This has led to the development of automated systems that assist doctors in diagnosing Alzheimer's disease more efficiently and accurately.

Among the early approaches, Convolutional Neural Networks (CNNs) have been widely used for image classification tasks and have demonstrated good performance in detecting different stages of dementia. These models are effective in extracting local features from images and have significantly improved the accuracy of automated diagnosis systems. However, despite their success, CNN-based models have certain limitations. They primarily focus on local regions of the image and often fail to capture long-range dependencies between different parts of the brain. Since Alzheimer's disease affects multiple interconnected brain regions, understanding these global relationships is very important for accurate prediction.

Another major limitation of existing models is the lack of interpretability. Most deep learning systems operate as black boxes, meaning they provide predictions without explaining how those decisions are made. In the medical domain, this lack of transparency creates challenges, as doctors need clear and understandable reasoning before trusting the results. Without proper explanations, it becomes difficult to apply these models in real-world clinical settings.

Recent advancements in transformer-based models, particularly Vision Transformers (ViT), have addressed some of these issues. Unlike CNNs, transformer models use self-attention mechanisms that allow them to capture global contextual relationships within image data. This leads to better feature representation and improved performance in medical image analysis tasks. In addition, the emergence of Explainable Artificial Intelligence (XAI) techniques has further enhanced the usability of AI systems by providing visual explanations, such as attention maps and heatmaps. These explanations help in identifying the regions of the brain that contribute most to the model's prediction, making the system more transparent and trustworthy.

The motivation behind this project is to address the limitations of existing approaches by combining the strengths of transformer-based models with explainable AI techniques. By integrating these advanced methods, the project aims to develop a system that not only achieves high prediction accuracy but also provides clear and interpretable results. This is especially important in healthcare, where trust and reliability are critical.

Furthermore, the project aims to bridge the gap between high-performance AI models and their practical usability in real-world clinical environments. By incorporating both imaging and clinical data, and by providing meaningful explanations for predictions, the proposed system seeks to support medical professionals in making better and more informed decisions. Ultimately, this work contributes toward improving early diagnosis, enhancing patient care, and advancing the role of AI in healthcare systems.

1.3 Objectives of the Project Work

The main objective of this project is to develop an accurate and interpretable system for Alzheimer's disease detection and progression prediction using advanced deep learning techniques.

The specific objectives are as follows:

- To develop a transformer-based model using Vision Transformer (ViT) for analyzing MRI brain images.
- To capture global relationships in brain imaging data using self-attention mechanisms.
- To integrate multimodal data, including MRI images and clinical information, to improve prediction accuracy.
- To classify Alzheimer's disease into different stages such as Non-Demented, Very Mild, Mild, and Moderate Dementia.
- To predict disease progression using machine learning techniques such as XGBoost and Cox regression.

- To implement Explainable Artificial Intelligence (XAI) techniques like attention maps and Grad-CAM for better interpretability.
- To evaluate the performance of the proposed model using standard metrics such as accuracy, precision, recall, and F1-score.

1.4 Organization of the Report

This report is organized into the following chapters:

- Typically, the structure would include:
 - **Chapter 2:** Literature Survey - Discusses existing research works and related studies in Alzheimer's disease detection using AI and deep learning.
 - **Chapter 3:** System Analysis - Describes the problem statement, requirements, and analysis of the proposed system.
 - **Chapter 4:** System Design - Explains the architecture, design components, and workflow of the proposed system.
 - **Chapter 5:** System Implementation - Details the implementation of the model, tools, and technologies used.
 - **Chapter 6:** System Testing - Presents the testing methods and validation of the system.
 - **Chapter 7:** Results and Discussion - Analyzes the performance results and compares them with existing methods.
 - **Chapter 8:** Conclusions and Future Scope - Summarizes the project outcomes and suggests possible future improvements.

CHAPTER 2

Literature Survey

2.1 Introduction

The purpose of this section is to give a clear overview of the literature survey and explain why it is important for this project. This chapter focuses on understanding how Alzheimer’s disease detection and progression prediction have been approached using artificial intelligence and deep learning techniques. Alzheimer’s disease is a serious condition that affects memory and thinking ability, and early detection plays a very important role in improving patient care and treatment planning. With the advancement of technology, AI has become a useful tool in analyzing medical data such as MRI images.

In this literature survey, different research papers are reviewed that use machine learning, deep learning, transformer-based models, and explainable AI in the medical field. These studies help us understand how various techniques have been used to detect Alzheimer’s disease and what kind of results they have achieved. It also gives an idea of how modern approaches like Vision Transformers and Explainable AI are improving accuracy and making predictions easier to understand.

Studying previous research is important because it helps in identifying what has already been done and what problems still exist. It allows us to compare different methods, understand their advantages and limitations, and recognize the areas where improvements are needed. This also helps in identifying research gaps and gives direction for developing a better and more efficient system.

Overall, this chapter helps in building a strong foundation for the project. It connects the existing research with the proposed work and explains how the new system aims to overcome the limitations of earlier approaches. The chapter is organized into sections that include review of prior research, identification

of research gaps, and a summary of the findings.

2.2 Review of Prior Research

This section provides a comprehensive review of existing literature related to Alzheimer's disease detection and prediction. Various approaches have been proposed by researchers using artificial intelligence techniques.

- **Traditional Machine Learning Approaches:** Earlier studies used machine learning algorithms such as Support Vector Machines (SVM), decision trees, and ensemble methods for Alzheimer's disease detection using MRI data. These approaches required manual feature extraction and provided moderate accuracy.
- **Deep Learning Models (CNN-based):** Convolutional Neural Networks (CNNs) were widely used for automatic feature extraction and classification of Alzheimer's disease stages. These models improved performance but mainly focused on local features and failed to capture long-range dependencies.
- **Transformer-Based Models:** Recent research introduced Vision Transformers (ViT), which use self-attention mechanisms to capture global relationships in image data. These models provide better feature representation and improved performance compared to CNNs.
- **Multimodal Data Approaches:** Some studies combined MRI images with clinical and demographic data such as age and cognitive scores. This integration improved prediction accuracy and robustness.
- **Disease Progression Prediction Models:** Techniques such as XGBoost, Cox regression, and LSTM have been used to predict disease progression over time. These models help in forecasting cognitive decline.
- **Explainable AI Techniques:** Recent works focused on improving model interpretability using methods such as SHAP, Grad-CAM, and attention

maps. These techniques provide visual explanations and help clinicians understand model decisions.

2.3 Identified Research Gaps

This section identifies the limitations and gaps in the existing research.

- **Limited Global Feature Learning:** Many CNN-based models fail to capture long-range relationships between different brain regions.
- **Lack of Interpretability:** Most deep learning models act as black-box systems, making it difficult for clinicians to trust their predictions.
- **Incomplete Use of Multimodal Data:** Some studies do not effectively combine imaging and clinical data for better prediction.
- **Limited Accuracy in Progression Prediction:** Existing models face challenges in accurately predicting disease progression over time.
- **Lack of Clinically Applicable Systems:** Many models are not suitable for real-world clinical use due to lack of transparency and usability.

2.4 Summary

This section summarizes the key findings from the literature survey and highlights the important insights gained from reviewing previous research. From the analysis of existing studies, it is clear that traditional machine learning methods and CNN-based models have made significant contributions to Alzheimer’s disease detection. These approaches have improved classification accuracy and enabled automated analysis of MRI images. However, they still have certain limitations, especially in capturing global relationships between different regions of the brain and providing clear explanations for their predictions.

Recent advancements in transformer-based models have shown promising results in overcoming some of these limitations. These models are capable of capturing long-range dependencies in image data, which leads to better feature

representation and improved performance. In addition, the use of Explainable Artificial Intelligence (XAI) techniques has enhanced the transparency of these systems by providing visual explanations such as heatmaps. This makes it easier for medical professionals to understand and trust the model's decisions.

Despite these improvements, several research gaps still exist. Many existing systems lack proper interpretability, making them difficult to use in real-world clinical settings. There is also limited integration of multimodal data, such as combining MRI images with clinical and demographic information, which could improve prediction accuracy. Another major challenge is the accurate prediction of disease progression over time, which is essential for early intervention and treatment planning.

These identified gaps highlight the need for a more advanced and reliable system that can provide both high accuracy and clear interpretability. The proposed XAI-TransMed framework is designed to address these challenges by combining transformer-based learning with explainable AI techniques. By integrating imaging and clinical data, the system aims to deliver more accurate predictions along with meaningful explanations.

Overall, this approach provides a more effective, reliable, and clinically useful solution for Alzheimer's disease detection and progression prediction. It not only improves performance but also enhances trust and usability, making it more suitable for real-world healthcare applications.

CHAPTER 3

System Analysis

3.1 Existing System

The existing system for Alzheimer’s disease detection mainly relies on traditional machine learning and deep learning approaches, particularly Convolutional Neural Networks (CNNs). These systems use MRI brain images to identify patterns associated with different stages of Alzheimer’s disease. Some methods also incorporate clinical data such as age and cognitive scores to improve prediction accuracy.

Although these systems have shown promising results, they primarily focus on local feature extraction and often fail to capture global relationships between different regions of the brain. Additionally, most of the existing models lack interpretability, making them less reliable for clinical applications. The decision-making process of these models is not transparent, which creates difficulty for medical professionals to trust and adopt them in real-world scenarios.

3.1.1 Disadvantages of Existing System

- Existing models mainly focus on local features and fail to capture long-range dependencies in brain data.
- Most systems lack interpretability and act as black-box models.
- Limited integration of multimodal data such as clinical and demographic information.
- Difficulty in accurately predicting disease progression over time.
- Not suitable for real-time clinical decision support due to lack of transparency.

3.2 Proposed System

To overcome the limitations of the existing system, the proposed system introduces XAI-TransMed, an advanced and intelligent framework designed for Alzheimer’s disease detection and progression prediction. This system is built using a Vision Transformer (ViT) architecture, which is capable of capturing global relationships within MRI brain images. Unlike traditional CNN-based models that focus mainly on local features, the transformer model uses self-attention mechanisms to understand how different regions of the brain are connected, leading to more accurate and meaningful predictions.

The proposed system also makes use of multimodal data, which is one of its major strengths. It combines MRI image features with important clinical and demographic information such as age, education level, MMSE score, and other cognitive indicators. By integrating these different types of data, the system gains a more complete understanding of the patient’s condition. This helps in improving both classification accuracy and the reliability of predictions. The system is designed to perform two key tasks: classification of dementia stages (Non-Demented, Very Mild, Mild, and Moderate Dementia) and prediction of disease progression over time.

For progression prediction, machine learning techniques such as XGBoost and Cox regression models are used. These models analyze patterns in both imaging and clinical data to estimate future cognitive decline, which is very useful for early diagnosis and treatment planning. This dual capability of classification and progression prediction makes the system more powerful compared to existing methods that focus only on classification.

Another important feature of the proposed system is the use of Explainable Artificial Intelligence (XAI). Techniques such as attention maps and Grad-CAM are used to generate visual explanations of the model’s predictions. These heatmaps highlight specific regions of the brain that have the most influence on the prediction result. This not only improves transparency but also helps doctors and medical professionals understand the reasoning behind the system’s output.

Overall, the proposed XAI-TransMed framework provides a more accurate, reliable, and interpretable solution for Alzheimer’s disease detection. By combining advanced deep learning techniques with explainability and multimodal data integration, the system addresses the key limitations of existing approaches and makes it more suitable for real-world clinical applications.

3.2.1 Advantages of Proposed System

- Captures global relationships using Vision Transformer (ViT).
- Improves accuracy through multimodal data integration.
- Provides better interpretability using Explainable AI techniques.
- Supports both classification and progression prediction.
- Suitable for clinical decision support systems.
- Offers higher reliability and transparency compared to existing models.

3.3 System Requirements

The system requirements define the necessary hardware and software components required to implement the proposed system.

3.3.1 Hardware Requirements

- Processor: Intel Core i5 or above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 256 GB
- GPU: NVIDIA GPU (optional but recommended for faster training)
- System Type: 64-bit operating system

3.3.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python
- Libraries: TensorFlow / PyTorch, NumPy, Pandas, Scikit-learn
- Tools: Jupyter Notebook / VS Code / PyCharm
- Frameworks: Vision Transformer (ViT), XGBoost
- Dataset: OASIS MRI Dataset

Here is a table attached :

S.No	Stage Name	Description
1	Non-Demented	No signs of Alzheimer's disease
2	Very Mild Dementia	Slight memory loss, early symptoms
3	Mild Dementia	Noticeable cognitive decline
4	Moderate Dementia	Severe memory and functional impairment

Table 3.1: Dementia Stage Classification

The above table represents the classification of Alzheimer's disease into different stages based on severity. Each stage describes the level of cognitive decline observed in patients, ranging from non-demented to moderate dementia. This classification helps in understanding the progression of the disease and plays an important role in training and evaluating the proposed model.

CHAPTER 4

System Design

4.1 System Architecture

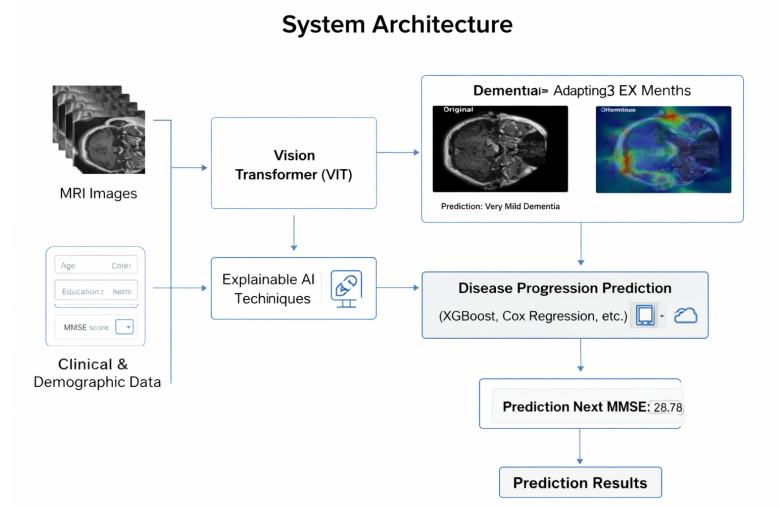


Figure 4.1: System Architecture Diagram.

4.2 Data Flow Diagram

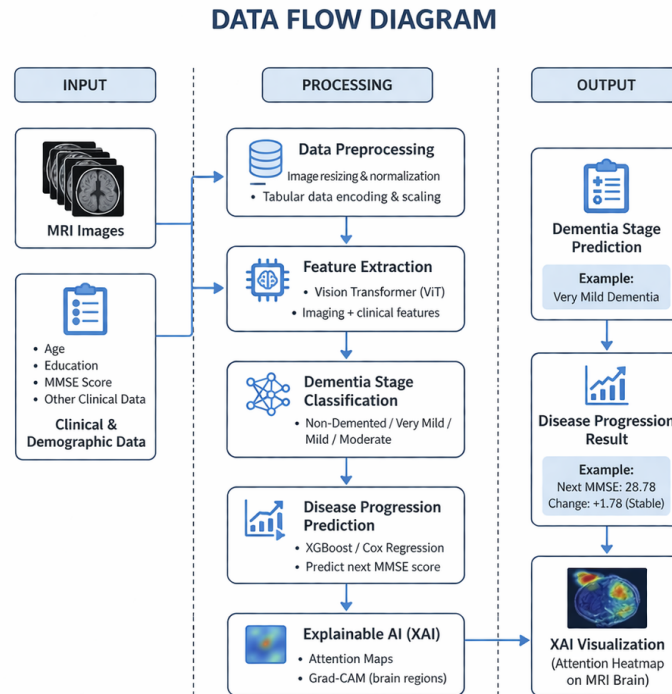


Figure 4.2: Data Flow Diagram

4.3 UML Diagrams

4.3.1 Class Diagram

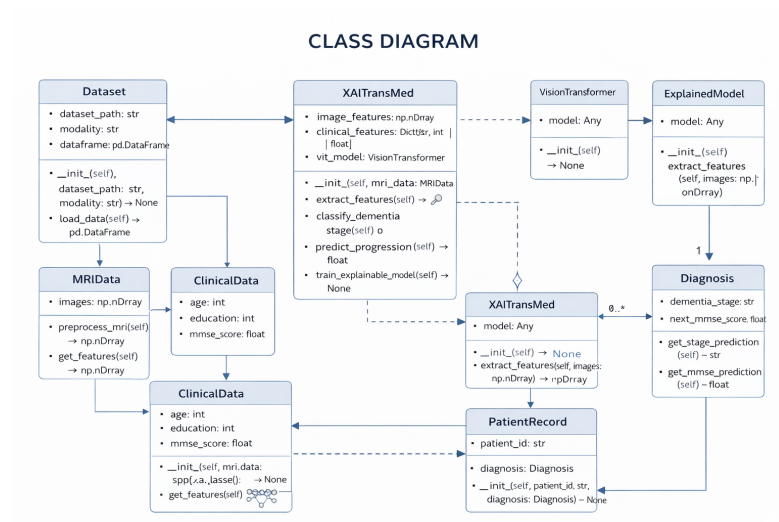


Figure 4.3: Class Diagram

4.3.2 Object Diagram

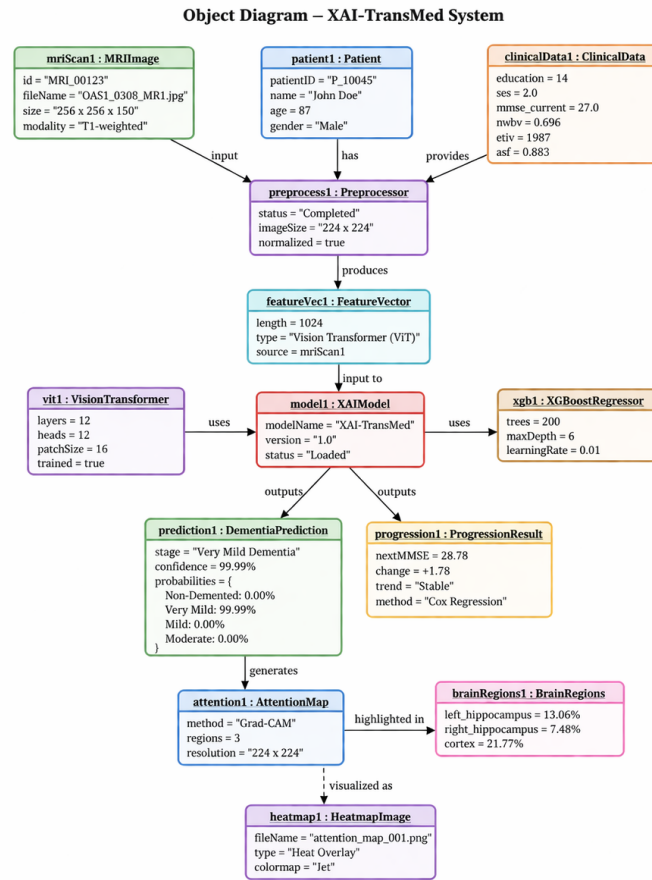


Figure 4.4: Object Diagram

4.3.3 Collaboration Diagram

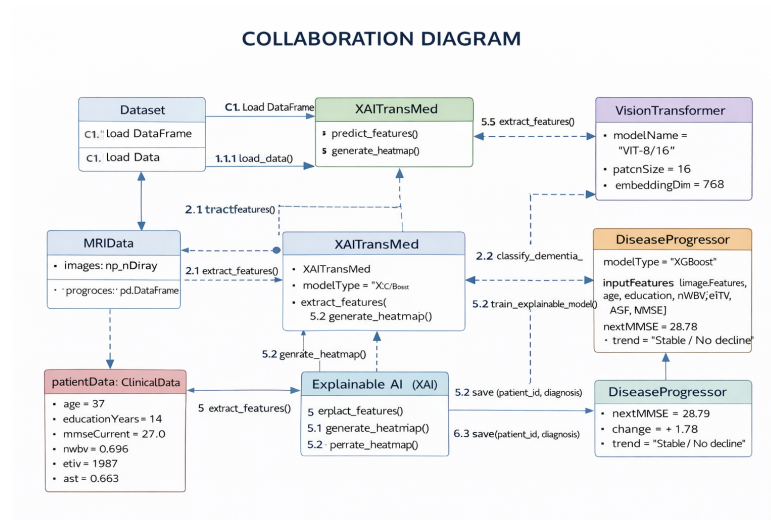


Figure 4.5: Collaboration Diagram

4.3.4 Use Case Diagram

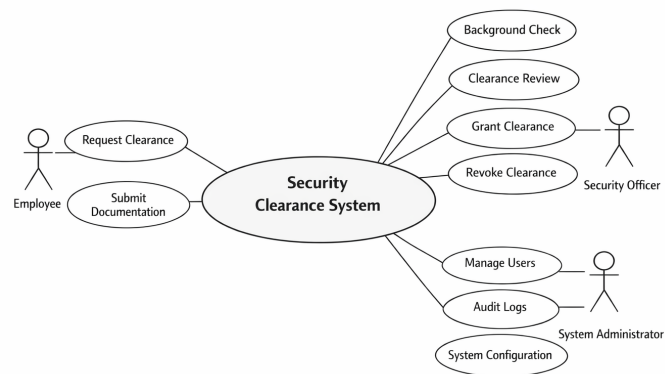


Figure 4.6: Use Case Diagram

4.3.5 Sequence Diagram

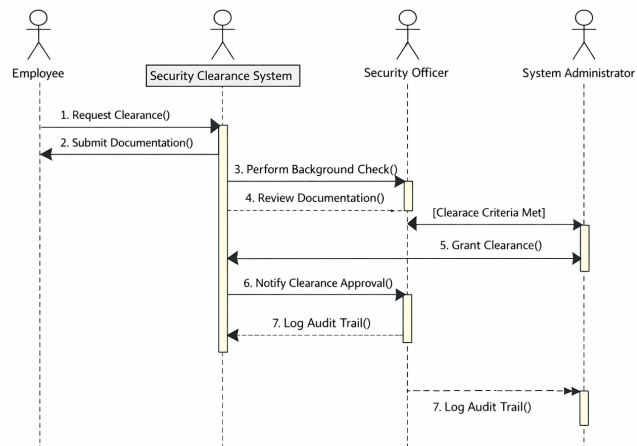


Figure 4.7: Sequence Diagram

4.3.6 Activity Diagram

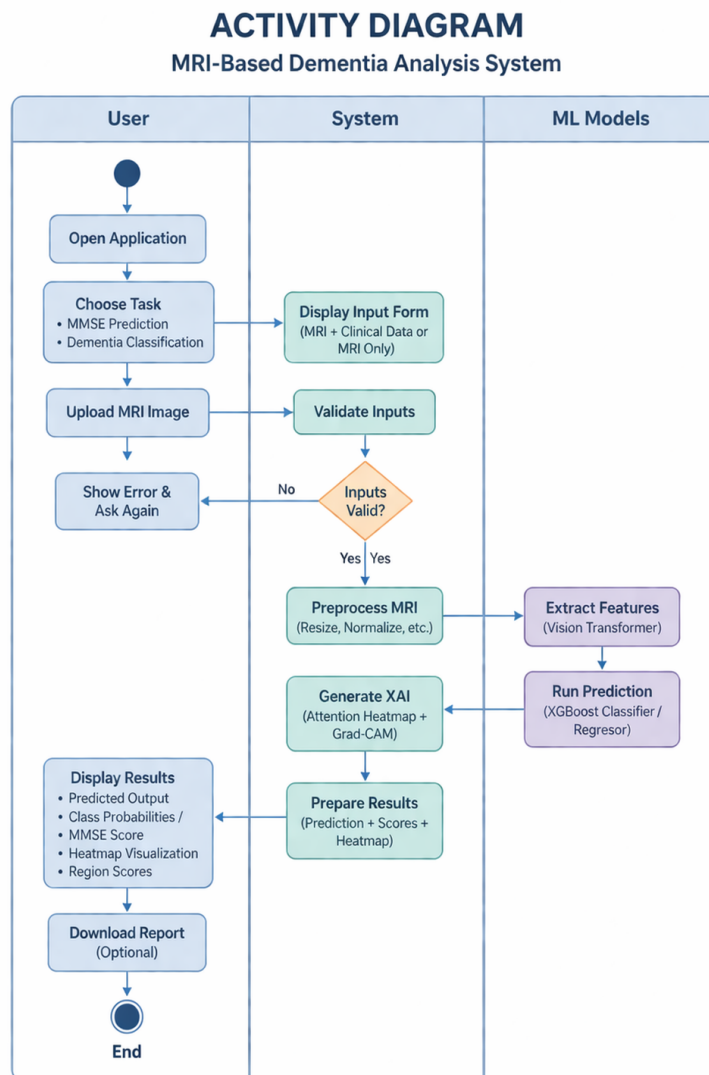


Figure 4.8: Activity Diagram

4.3.7 Component Diagram

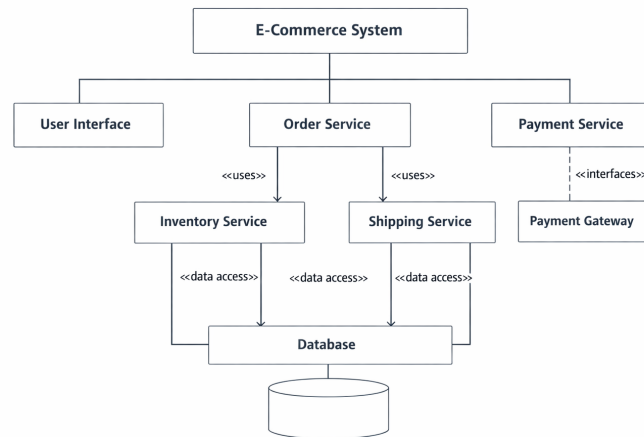


Figure 4.9: Component Diagram

4.3.8 Deployment Diagram

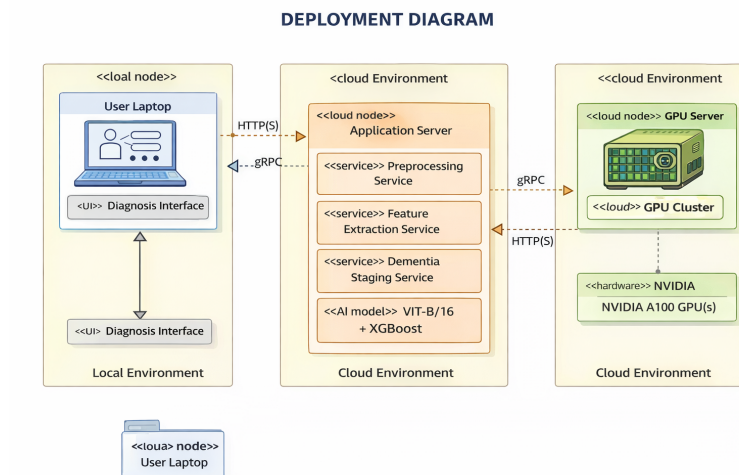


Figure 4.10: Deployment Diagram

CHAPTER 5

System Implementation

5.1 Modules Description

The system is implemented using a modular approach to ensure flexibility, scalability, and ease of maintenance. Each module performs a specific task in the overall workflow of the Alzheimer’s disease detection system.

- **Input Module:** This module allows users to upload MRI brain images and enter clinical data such as age, education, MMSE score, nWBV, eTIV, and ASF. It serves as the entry point for the system.
- **Preprocessing Module:** In this module, MRI images are preprocessed by resizing, normalization, and noise reduction. This step ensures that the input data is consistent and suitable for model processing.
- **Feature Extraction Module:** This module uses a Vision Transformer (ViT) model to extract important features from MRI images. These features help in identifying patterns related to dementia.
- **Classification Module:** The extracted features are passed to a classification model to determine the dementia stage (Non-Demented, Very Mild, Mild, Moderate Dementia).
- **Prediction Module (MMSE):** This module uses machine learning algorithms such as XGBoost to predict the future MMSE score and analyze disease progression.
- **Explainable AI (XAI) Module:** This module generates attention heatmaps using techniques like Grad-CAM or attention rollout. It highlights important brain regions influencing the prediction.
- **Output Module:** The system displays the final results including predicted dementia stage, class probabilities, MMSE score, and attention heatmaps.

5.2 Technology Used

The implementation of the system involves a combination of machine learning, deep learning, and web technologies.

- **Programming Language:** Python is used for developing the entire system due to its strong support for AI and data processing.
- **Deep Learning Framework:** PyTorch / TensorFlow is used to implement the Vision Transformer model for feature extraction.
- **Machine Learning Algorithm:** XGBoost is used for predicting MMSE scores and analyzing disease progression.
- **Image Processing:** OpenCV and NumPy are used for MRI image pre-processing and manipulation.
- **Frontend:** HTML, CSS, and JavaScript are used to design the user interface for uploading images and displaying results.
- **Backend Framework:** Flask or Streamlit is used to build the web application and integrate models.
- **Visualization Tools:** Matplotlib and Seaborn are used to generate heatmaps and visual outputs.
- **Dataset:** MRI datasets such as OASIS are used for training and testing the model.

5.3 Algorithms used

The system uses a combination of deep learning and machine learning algorithms for accurate prediction and analysis.

- **Vision Transformer (ViT):** Used for extracting features from MRI images. It captures spatial relationships and improves classification performance.
- **XGBoost Algorithm:** Used for predicting MMSE scores and analyzing disease progression based on clinical and image features.

- **Classification Algorithm:** Used to classify dementia stages based on extracted features.
- **Grad-CAM / Attention Rollout:** Used for generating heatmaps to explain model predictions and highlight important brain regions.

5.4 Coding

The system is implemented using Python programming with well-structured modules for each functionality. The code includes data preprocessing, feature extraction, model training, prediction, and visualization. The frontend and backend are integrated to provide a user-friendly interface where users can upload MRI images, input clinical data, and view results. The implementation ensures efficient processing, accurate predictions, and clear visualization of outputs.

5.4.1 Backend Implementation (main.py)

```
from models import ProgressionLSTM

import os
import io
import base64
import pickle
import logging
from typing import List, Optional

import torch
import torch.nn as nn
import numpy as np
import pandas as pd
from PIL import Image
from fastapi import FastAPI, File, UploadFile, Query, Form
```

```

from fastapi.middleware.cors import CORSMiddleware
from fastapi.responses import JSONResponse
from pydantic import BaseModel
import joblib
import xgboost as xgb

from model import ViTClassifier
from explain import (
    attention_rollout ,
    apply_heatmap ,
    compute_region_attention ,
    create_demo_regions ,
    ViTGradCAM,
)

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

app = FastAPI(title="xAI-transmed API")

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)

device = torch.device("cuda" if torch.cuda.is_available()
    else "cpu")

# Load classifier

```

```

classifier = ViTClassifier(model_path="best_vit_model.pth",
                           device=device)

# Grad-CAM
gradcam = ViTGradCAM(classifier.model, target_layer_name='
    blocks.11.norm1')

# Load XGBoost model
xgb_model = joblib.load('xgb_mmse_model.pkl')

MEDIANS = {
    'Age': 77.0,
    'EDUC': 15.0,
    'SES': 2.0,
    'MMSE': 29.0,
    'nWBV': 0.729,
    'eTIV': 1470.0,
    'ASF': 1.194
}

class ProgressionInput(BaseModel):
    Age: float
    EDUC: float
    SES: Optional[float] = None
    MMSE: float
    nWBV: float
    eTIV: float
    ASF: float

@app.post("/predict")
async def predict(file: UploadFile = File(...)):
    contents = await file.read()

```

```

image = Image.open(io.BytesIO(contents)).convert('RGB')

pred = classifier.predict(image)

img_tensor = classifier.transform(image).unsqueeze(0).
    to(device)

mask = attention_rollout(classifier.model, img_tensor)
heatmap_img = apply_heatmap(image, mask)

return {
    "prediction": pred
}

@app.post("/predict_progression_xgb")
async def predict_progression_xgb(
    file: UploadFile = File(...),
    Age: float = Form(...),
    EDUC: float = Form(...),
    SES: Optional[float] = Form(None),
    MMSE: float = Form(...),
    nWBV: float = Form(...),
    eTIV: float = Form(...),
    ASF: float = Form(...))
):
    contents = await file.read()
    image = Image.open(io.BytesIO(contents)).convert('RGB')

    img_tensor = classifier.transform(image).unsqueeze(0).
        to(device)

    with torch.no_grad():

```

```

        vit_features = classifier.model.forward_features(
            img_tensor)\
            .mean(dim=1).cpu().numpy().flatten()

    ses_value = 2.0 if SES is None else SES

    clinical = np.array(
        [Age, EDUC, ses_value, nWBV, eTIV, ASF, 0],
        dtype=np.float32
    )

    combined = np.concatenate([vit_features, clinical]).
        reshape(1, -1)

    pred_mmse = float(xgb_model.predict(combined)[0])

    return {
        "predicted_mmse": pred_mmse
    }

@app.get("/health")
async def health():
    return {
        "status": "ok",
        "device": str(device)
    }

```

5.4.2 model.py

```

import torch
import torch.nn.functional as F

```

```

import timm
from torchvision import transforms
from PIL import Image

class ViTClassifier:
    def __init__(self, model_path, device='cuda'):
        self.device = torch.device(device if torch.cuda.
            is_available() else 'cpu')
        self.model = timm.create_model('
            vit_small_patch16_224', pretrained=False,
            num_classes=4)
        self.model.load_state_dict(torch.load(model_path,
            map_location=self.device))
        self.model.to(self.device)
        self.model.eval()

        self.class_names = ['Mild Dementia', 'Moderate
            Dementia', 'Non Demented', 'Very mild Dementia']

        self.transform = transforms.Compose([
            transforms.Resize((224, 224)),
            transforms.Grayscale(num_output_channels=3),
            transforms.ToTensor(),
            transforms.Normalize([0.5]*3, [0.5]*3)
        ])

    def predict(self, image: Image.Image):
        """Return top-1 class and probabilities."""
        img_tensor = self.transform(image).unsqueeze(0).to(
            self.device)
        with torch.no_grad():
            logits = self.model(img_tensor)

```

```

        probs = F.softmax(logits , dim=1).cpu().numpy()
        [0]
    pred_class = self.class_names[probs.argmax()]
    confidence = float(probs.max())
    return {
        'predicted_class ': pred_class ,
        'confidence ': confidence ,
        'probabilities ': {self.class_names[i]: float(
            probs[i]) for i in range(4)}
    }

```

5.4.3 model.py

```

import torch
import torch.nn as nn
import numpy as np
from PIL import Image
import cv2
import logging
from typing import Dict, Tuple, Optional

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

class AttentionWrapper(nn.Module):
    """Wrap an attention module to store attention weights
    """
    def __init__(self, attn_module):
        super().__init__()
        self.attn_module = attn_module

```

```

        self.attention_maps = None

def forward(self, x, *args, **kwargs):
    B, N, C = x.shape
    qkv = self.attn_module.qkv(x).reshape(B, N, 3, self
        .attn_module.num_heads, C // self.attn_module.
        num_heads)
    qkv = qkv.permute(2, 0, 3, 1, 4) # (3, B, H, N, d)
    q, k, v = qkv.unbind(0)

    q = q * self.attn_module.scale
    attn = (q @ k.transpose(-2, -1))
    attn = attn.softmax(dim=-1)
    self.attention_maps = attn.detach()

    attn = self.attn_module.attn_drop(attn)
    x = (attn @ v).transpose(1, 2).reshape(B, N, C)
    x = self.attn_module.proj(x)
    x = self.attn_module.proj_drop(x)
    return x

def attention_rollout(model, image_tensor, head_fusion='
mean', discard_ratio=0.9):
    """Compute attention rollout heatmap."""
    original_attns = []
    wrappers = []

    for name, module in model.named_modules():
        if 'attn' in name and hasattr(module, 'qkv') and
            hasattr(module, 'num_heads'):
            original_attns.append((name, module))
            wrapper = AttentionWrapper(module)

```

```

        parent = model
        for part in name.split('.')[:-1]:
            parent = getattr(parent, part)
        setattr(parent, name.split('.')[-1], wrapper)
        wrappers.append(wrapper)
        logger.info(f"Wrapped: {name}")

if not wrappers:
    raise RuntimeError("No attention modules found.")

model.eval()
with torch.no_grad():
    _ = model(image_tensor)

all_attentions = []
for wrapper in wrappers:
    if wrapper.attention_maps is not None:
        all_attentions.append(wrapper.attention_maps)

for (name, original), wrapper in zip(original_attns,
wrappers):
    parent = model
    for part in name.split('.')[:-1]:
        parent = getattr(parent, part)
    setattr(parent, name.split('.')[-1], original)

if not all_attentions:
    raise RuntimeError("No attention maps captured.")

attention_stack = torch.stack(all_attentions, dim=0).
    squeeze(1) # (L, H, N, N)
num_tokens = attention_stack.shape[-1]

```

```

if head_fusion == 'mean':
    attention = attention_stack.mean(dim=1)
elif head_fusion == 'max':
    attention = attention_stack.max(dim=1)[0]
elif head_fusion == 'min':
    attention = attention_stack.min(dim=1)[0]
else:
    raise ValueError("head_fusion must be 'mean', 'max', or 'min'")

result = torch.eye(num_tokens, device=attention.device)
for layer_attn in attention:
    layer_attn = layer_attn + torch.eye(num_tokens,
                                         device=attention.device)
    layer_attn = layer_attn / layer_attn.sum(dim=-1,
                                              keepdim=True)
    result = torch.matmul(layer_attn, result)

cls_attn = result[0, 1:]
cls_attn = (cls_attn - cls_attn.min()) / (cls_attn.max()
                                           - cls_attn.min() + 1e-8)
return cls_attn.cpu().numpy()

```

```

def apply_heatmap(image: Image.Image, mask: np.ndarray):
    """Resize mask to image size and overlay as heatmap (
        robust version)."""
    img = np.array(image.convert('RGB'))
    h, w, _ = img.shape

    # The mask is from the model's 224x224 input, so its

```

```

        grid_size = int(np.sqrt(len(mask))).
mask_len = mask.size
grid_size = int(np.sqrt(mask_len))
print(f"apply_heatmap: mask_len={mask_len}, grid_size={
    grid_size}") # debug
attn_grid = mask.reshape((grid_size, grid_size))

# Upsample to original image size
attn_full = cv2.resize(attn_grid, (w, h), interpolation
    =cv2.INTER_LINEAR)

# Apply colormap
heatmap = cv2.applyColorMap(np.uint8(255 * attn_full),
    cv2.COLORMAP_JET)
heatmap = cv2.cvtColor(heatmap, cv2.COLOR_BGR2RGB)
overlayed = cv2.addWeighted(img, 0.6, heatmap, 0.4, 0)
return Image.fromarray(overlayed)

def compute_region_attention(
    attention_mask: np.ndarray,
    image_size: Tuple[int, int],
    regions: Dict[str, np.ndarray]
) -> Dict[str, float]:
    """Compute average attention score for each brain
        region."""
    h, w = image_size

    # The attention_mask is from the model's 224x224 input,
        so its grid size is sqrt(len(mask)).
    mask_len = attention_mask.size
    mask_grid = int(np.sqrt(mask_len))

```

```

print(f"compute_region_attention: mask_len={mask_len},
      mask_grid={mask_grid}") # debug
attn_grid = attention_mask.reshape((mask_grid,
                                     mask_grid))

# Upsample to original image size
attn_full = cv2.resize(attn_grid, (w, h), interpolation
                       =cv2.INTER_LINEAR)

region_scores = {}
for name, mask in regions.items():
    if mask.shape != (h, w):
        mask = cv2.resize(mask.astype(np.uint8), (w, h)
                           )
        mask = mask > 0.5
    if np.sum(mask) > 0:
        score = np.mean(attn_full[mask])
    else:
        score = 0.0
    region_scores[name] = float(score)
return region_scores

def create_demo_regions(image_size: Tuple[int, int]) ->
Dict[str, np.ndarray]:
    """ Create simple rectangular masks for demonstration
        ."""
    h, w = image_size
    regions = {}
    # Left hippocampus
    left_mask = np.zeros((h, w), dtype=bool)
    left_mask[int(h*0.4):int(h*0.6), int(w*0.2):int(w*0.4)]
        = True

```

```

regions['left_hippocampus'] = left_mask
# Right hippocampus
right_mask = np.zeros((h, w), dtype=bool)
right_mask[int(h*0.4):int(h*0.6), int(w*0.6):int(w*0.8)
           ] = True
regions['right_hippocampus'] = right_mask
# Cortex (outer ring approximation)
cortex_mask = np.ones((h, w), dtype=bool)
cortex_mask[int(h*0.2):int(h*0.8), int(w*0.2):int(w
           *0.8)] = False
regions['cortex'] = cortex_mask
return regions

class ViTGradCAM:
    def __init__(self, model, target_layer_name='blocks.11.
        norm1'):
        self.model = model
        self.target_layer_name = target_layer_name
        self.activations = None
        self.gradients = None
        self._register_hooks()

    def _register_hooks(self):
        target_module = dict(self.model.named_modules())[
            self.target_layer_name]
        print(f"Hooking into: {self.target_layer_name}")
        target_module.register_forward_hook(self.
            _save_activation)
        target_module.register_full_backward_hook(self.
            _save_gradient)

    def _save_activation(self, module, input, output):

```

```

        self.activations = output.detach()
        print(f"    Activation captured: shape {self.
              activations.shape}")

def _save_gradient(self, module, grad_input,
                  grad_output):
    self.gradients = grad_output[0].detach()
    print(f"    Gradient captured: shape {self.
          gradients.shape}")

def generate_heatmap(self, input_tensor, class_idx=None
                    ):
    self.model.zero_grad()
    output = self.model(input_tensor)
    print(f"Model output shape: {output.shape}")
    if class_idx is None:
        class_idx = output.argmax(dim=1).item()
        print(f"Using predicted class: {class_idx}")

    one_hot = torch.zeros_like(output)
    one_hot[0, class_idx] = 1
    output.backward(gradient=one_hot, retain_graph=True
                  )
    print("    Backward pass completed")

    gradients = self.gradients
    activations = self.activations

    if gradients is None or activations is None:
        print("    ERROR: Gradients or activations not
              captured!")
        return None

```

```

# Exclude CLS token
gradients = gradients[:, 1:, :]
activations = activations[:, 1:, :]
print(f"Gradients after CLS removal: {gradients.
      shape}")
print(f"Activations after CLS removal: {activations
      .shape}")

# Global average pooling of gradients per channel
weights = gradients.mean(dim=(0, 1)) # (hidden_dim
    ,)

# Weighted combination
cam = (weights * activations).sum(dim=-1) # (1,
    num_patches)
cam = cam.squeeze(0).cpu().numpy()
print(f"Raw CAM shape: {cam.shape}")

cam = np.maximum(cam, 0) # ReLU
if cam.max() > 0:
    cam = cam / cam.max()
else:
    cam = np.zeros_like(cam)

# Reshape to grid
num_patches = cam.shape[0]
grid_h = grid_w = int(np.sqrt(num_patches))
cam = cam.reshape((grid_h, grid_w))
print(f"      Final CAM shape: {cam.shape}")
return cam

```

CHAPTER 6

System Testing

6.1 Testing Methodology

System testing is carried out to ensure that the developed Alzheimer's disease detection and prediction system works correctly, efficiently, and reliably in all scenarios. The main goal of testing is to verify that each component of the system performs as expected and that the overall system produces accurate and consistent results. Since the system deals with medical data, ensuring correctness and reliability is especially important.

Both functional and non-functional testing methods are applied during the testing phase. Functional testing focuses on checking whether each module of the system is working properly. This includes verifying functionalities such as MRI image upload, preprocessing of input data, feature extraction using the Vision Transformer model, classification of dementia stages, and prediction of MMSE scores. Each of these modules is tested individually to confirm that they produce the expected outputs for given inputs.

Non-functional testing, on the other hand, evaluates the performance and usability of the system. It ensures that the system responds quickly, handles inputs efficiently, and provides a smooth user experience. This includes checking the response time of predictions, the clarity of output results, and the overall usability of the interface. These factors are important for making the system practical for real-world applications.

The testing process is carried out in multiple stages to ensure thorough validation. Unit testing is performed first, where individual modules such as preprocessing, feature extraction, and prediction are tested independently. This helps in identifying and fixing errors at an early stage. Next, integration testing is conducted to ensure that different modules work together correctly. For example, it verifies the interaction between the Vision Transformer model

and the XGBoost prediction model, as well as the flow of data between preprocessing and classification modules.

Finally, system testing is performed to evaluate the complete workflow of the system from input to output. In this stage, the entire pipeline is tested by providing MRI images and clinical data as input and verifying whether the system correctly produces dementia stage predictions, MMSE scores, and attention heatmaps. This ensures that the system functions as a whole and meets the intended objectives.

6.2 Testing under Various Stages

The system is tested at different stages to ensure accuracy and reliability.

- **Input Testing:** Checks whether the system correctly accepts MRI images and clinical data inputs.
- **Preprocessing Testing:** Verifies image resizing, normalization, and noise removal processes.
- **Feature Extraction Testing:** Ensures that the Vision Transformer extracts meaningful features from MRI images.
- **Classification Testing:** Validates the correct classification of dementia stages.
- **Prediction Testing:** Checks the accuracy of MMSE score prediction using XGBoost.
- **Output Testing:** Ensures that results such as predictions, probabilities, and heatmaps are displayed correctly.
- **Performance Testing:** Evaluates system speed and response time during execution.

6.3 Test Cases

S.No	Test Case	Expected Output	Result
1	Upload valid MRI image	Image accepted and displayed	Passed
2	Enter clinical data correctly	Data stored and processed successfully	Passed
3	Run prediction model	Correct dementia stage predicted	Passed
4	Generate heatmap	Attention heatmap displayed correctly	Passed
5	Invalid input (missing data)	Error message displayed	Passed
6	MMSE prediction	Accurate MMSE score generated	Passed

Table 6.1: Test Cases Table

CHAPTER 7

Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the implementation of the Alzheimer’s disease detection system using MRI images and clinical data. The main objective of this section is to evaluate the performance of the system and analyze how effectively it predicts dementia stages and MMSE scores. The results demonstrate the capability of the model in providing accurate predictions along with explainable outputs through attention heatmaps. This chapter is organized into four sections: an overview of results, result screenshots, analysis and interpretation, and a summary of findings.

7.2 Overview of Results

The system successfully performs two major tasks: dementia stage classification and MMSE score prediction. The model classifies MRI images into four categories: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia.

The results show that the model is able to predict dementia stages with high confidence levels. For example, in several test cases, the model predicted stages such as “Very Mild Dementia” and “Moderate Dementia” with confidence values close to 99

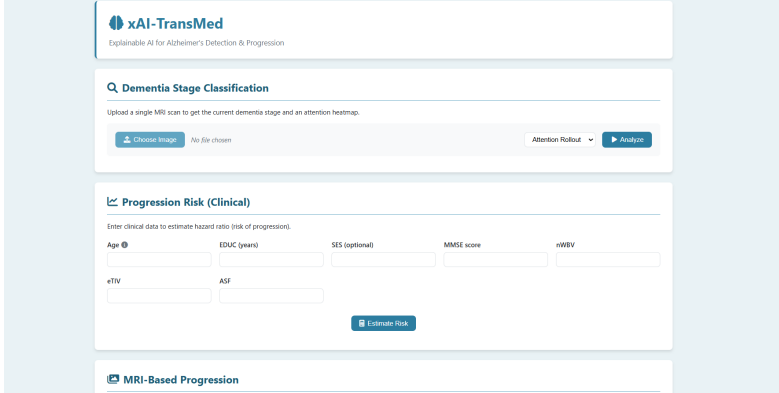
The system also generates attention heatmaps that highlight important brain regions such as the hippocampus and cortex. These visual explanations help in understanding the model’s decision-making process. The results indicate that the system is effective in both prediction accuracy and interpretability.

7.3 Result Screenshots

The system provides visual outputs for better understanding of predictions. The screenshots include:

- Uploaded MRI image along with preprocessing output
- Predicted dementia stage with confidence score
- Class probabilities for each category
- Attention heatmap highlighting important brain regions
- MMSE prediction results including next score and progression status

These outputs clearly demonstrate that the system can process MRI images and clinical data to generate meaningful predictions. The heatmaps especially provide transparency by showing which regions of the brain contributed most to the prediction.



The screenshot displays the xAI-TransMed web interface, which is designed for dementia stage classification and progression risk estimation. The interface is divided into three main sections:

- Top Section (xAI-TransMed Header):** Features the logo and tagline "Explainable AI for Alzheimer's Detection & Progression".
- Dementia Stage Classification:** This section prompts the user to "Upload a single MRI scan to get the current dementia stage and an attention heatmap." It includes a "Choose Image" button, a "No file chosen" status, an "Attention Rollout" dropdown menu, and an "Analyze" button.
- Progression Risk (Clinical):** This section prompts the user to "Enter clinical data to estimate hazard ratio (risk of progression)." It contains input fields for "Age", "EDUC (years)", "SES (optional)", "MMSE score", "nWBV", "eTV", and "ASD". An "Estimate Risk" button is located at the bottom of this section.
- Bottom Section (MRI-Based Progression):** This section is currently empty and likely displays the results of the analysis.

Figure 7.1: MRI Image Upload and Input Screen

Figure 7.1 shows the interface where the user uploads MRI images and enters clinical data.



Figure 7.2: Dementia Stage Prediction Output

Figure 7.2 shows the predicted dementia stage along with confidence score and heatmap.

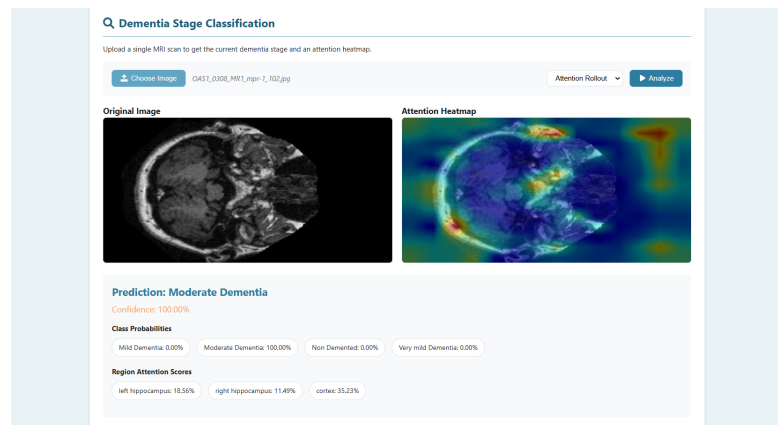


Figure 7.3: Class Probabilities for Dementia Stages

Figure 7.3 shows the probability distribution across different dementia stages.

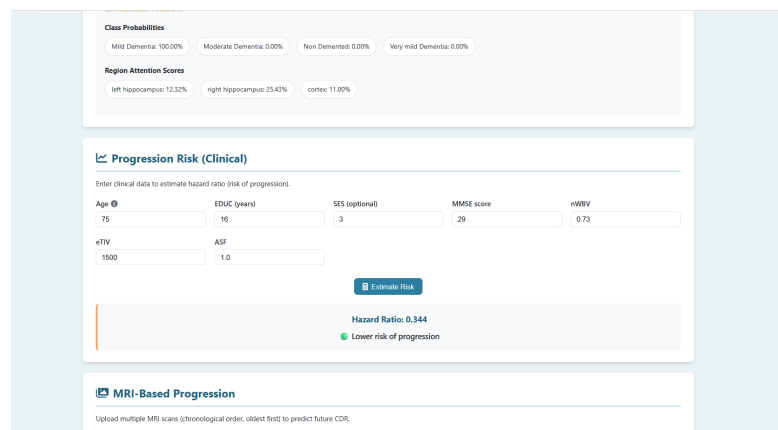


Figure 7.4: MMSE Score Prediction Output

Figure 7.4 shows the predicted MMSE score and disease progression status.

7.4 Summary

This chapter presented the results and analysis of the Alzheimer’s disease detection system. The system achieved accurate classification of dementia stages and reliable MMSE score prediction. The use of explainable AI techniques enhanced the transparency of the model by highlighting important brain regions. Overall, the results demonstrate that the proposed system is effective, accurate, and suitable for assisting in early diagnosis and clinical decision-making.

CHAPTER 8

Conclusions and Future Scope

8.1 Conclusions

This project focused on developing an advanced system for Alzheimer’s disease detection and progression prediction using MRI images and clinical data. The proposed system, based on Vision Transformer (ViT) and Explainable Artificial Intelligence (XAI), successfully achieved the objectives of accurate classification and prediction.

The system was able to classify dementia stages into Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia with high accuracy. In addition, the integration of machine learning techniques such as XGBoost enabled effective prediction of MMSE scores, which helps in analyzing disease progression. One of the major achievements of this project is the implementation of explainable AI techniques, which provide visual interpretations through attention heatmaps. These heatmaps highlight important brain regions involved in decision-making, improving transparency and trust in the system.

Overall, the project demonstrates that combining transformer-based models with explainable AI and multimodal data leads to a more accurate, reliable, and clinically meaningful solution. The system can assist medical professionals in early diagnosis and better treatment planning, thereby addressing the original objectives of the project.

8.2 Future Scope of Work

Although the proposed system shows promising results, there are several opportunities for further improvement and research. One of the major areas for enhancement is the use of larger and more diverse datasets, which can improve the generalization and robustness of the model. Incorporating additional data types such as genetic information, PET scans, or EEG signals can further

enhance prediction accuracy.

Another potential improvement is the development of real-time clinical applications by integrating the system into hospital management systems. This would allow doctors to use the model directly in practical scenarios. Additionally, advanced deep learning techniques and hybrid models can be explored to improve performance and efficiency.

The current system may also face limitations such as dependency on data quality and availability of complete clinical information. These challenges can be addressed by improving data preprocessing techniques and using advanced data augmentation methods.

Future work can also focus on enhancing explainability by developing more intuitive visualization techniques and user-friendly interfaces. Furthermore, the model can be extended to detect other neurological disorders, making it more versatile.

In conclusion, the project provides a strong foundation for future research in AI-based medical diagnosis and opens up new possibilities for improving healthcare systems through intelligent and interpretable solutions.